

Crystals and Symmetries

Representation theory of Point Groups and Space Groups

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- Bradley and Cracknell (1972): The Mathematical Theory of Symmetries in Solids
- Sands (1969): Introduction to Crystallography
- Moore (2014): Quantum Symmetries

- Bravais lattice and Point groups
- Space groups
- Representation theory of Space groups
- Bandstructure and consequences of symmetries
- Current frontiers and Summary

Bravais Lattice and Point Groups

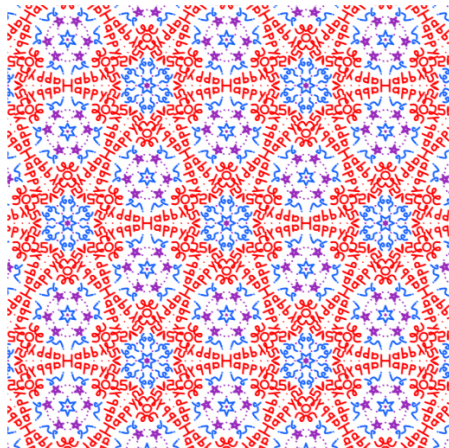
What is a Bravais Lattice?

A **Bravais lattice** is an infinite periodic array of points:

$$\mathbf{t} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3, \quad n_i \in \mathbb{Z}$$

It has two types of symmetry:

- **Translations:** shift by any lattice vector
- **Point operations:** rotations, reflections, inversions fixing the origin

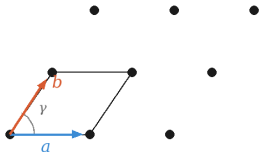


Source: www.xtal.iqf.csic.es/Cristalografia

2D Bravais lattices

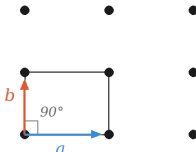
Oblique

$a \neq b$, $\gamma \neq 90^\circ, 120^\circ$



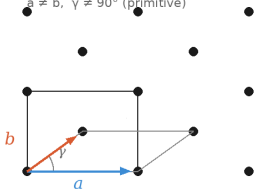
Rectangular

$a \neq b$, $\gamma = 90^\circ$



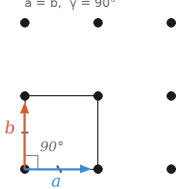
Centered rectangular

$a \neq b$, $\gamma = 90^\circ$ (primitive)



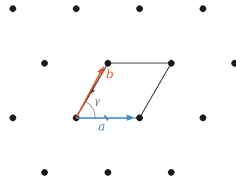
Square

$a = b$, $\gamma = 90^\circ$



Hexagonal

$a = b$, $\gamma = 120^\circ$



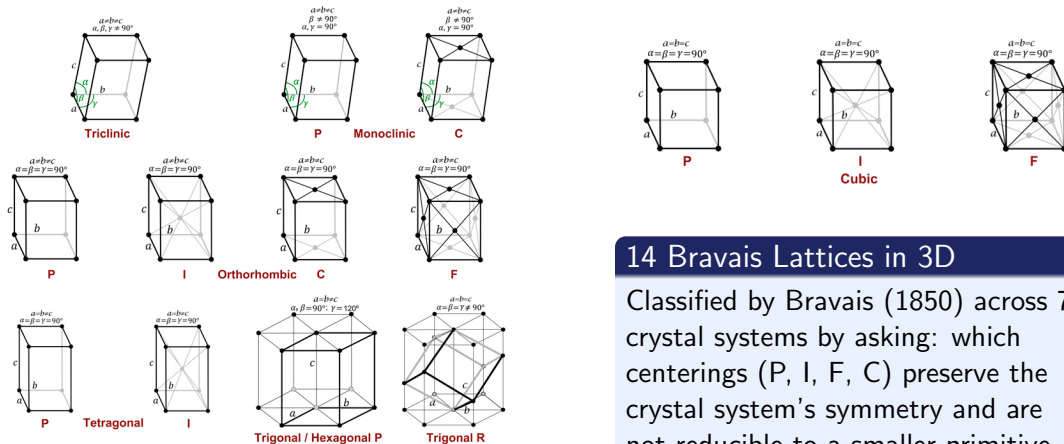
In different dimensions?

In $d=2$, 5 Bravais lattice

In $d=3$, 14 Bravais lattice

In $d=4$, 64 Bravais lattices

3D Bravais lattices



14 Bravais Lattices in 3D

Classified by Bravais (1850) across 7 crystal systems by asking: which centerings (P, I, F, C) preserve the crystal system's symmetry and are not reducible to a smaller primitive cell?

Source: www.xtal.iqf.csic.es/Cristalografia

Point Groups

- Symmetry operations in 3D Euclidean space, which keeping a point fixed, don't change lengths and angles, can be expressed by the group $O(3)$.
- For a 3×3 real matrix $R \in O(3)$, the vector $\mathbf{r}$ gets moved to $\mathbf{r}'$ as, $\mathbf{r}' = R\mathbf{r}$
 - **Proper** rotations only: $SO(3)$, $\det R = +1$
 - **Improper** operations: $\det R = -1$ (mirrors, inversions, roto-inversions)
- Every finite subgroup of $O(3)$ is a **point group**.

Key point

A *crystallographic point group* must satisfy the extra requirement that it should be compatible with the translational symmetry of the crystalline solid. The possible point group operations that may satisfy this extra requirement are:

E (*identity*), I (*inversion*), σ_z (*reflection*), $C_{nz}^{\pm}$ (*rotation by $\frac{2\pi}{n}$*) with $n \in \{1, 2, 3, 4, 6\}$.

Crystallographic Restriction Theorem

Claim: In 2 or 3 dimensions, a $\theta = \frac{2\pi}{n}$ rotation compatible with a lattice can only have order $n \in \{1, 2, 3, 4, 6\}$.

Proof:

- ① R must map lattice vectors to lattice vectors
- ② $\Rightarrow R$ has integer matrix entries in lattice basis
- ③ $\Rightarrow \text{Tr}(R) = 1 + 2 \cos \theta \in \mathbb{Z}$

Allowed values

$2 \cos \theta$	θ	Order n
2	0°	1
1	60°	6
0	90°	4
-1	120°	3
-2	180°	2

No 5-fold, 7-fold, ... symmetry!

The 7 Crystal Systems and 14 Bravais Lattices

Crystal System	# Bravais Lattices	# Point Groups
Triclinic	1	2
Monoclinic	2	3
Orthorhombic	4	3
Tetragonal	2	7
Trigonal	1	5
Hexagonal	1	7
Cubic	3	5
Total	14	32

Key point

The crystal system constrains *which point groups Q are allowed*; the specific Bravais lattice determines the translation subgroup T .

Space Groups

Seitz Notation $\{R | \mathbf{v}\}$

Every space group element is written as a **Seitz operator**:

$$\{R | \mathbf{v}\} \text{ acts on } \mathbf{x} \text{ as } \mathbf{x} \mapsto R\mathbf{x} + \mathbf{v}$$

Multiplication rule:

$$\{R_1 | \mathbf{v}_1\} \{R_2 | \mathbf{v}_2\} = \{R_1 R_2 | \mathbf{v}_1 + R_1 \mathbf{v}_2\}$$

Covers all symmetry operations

Operation	R	$\mathbf{v}$
Pure translation	id. matrix, E	lattice vector
Rotation	rot. matrix, $\det = +1$	lattice vector
Screw axis	rot. matrix, $\det = +1$	fractional
Mirror / inversion	$\det = -1$	lattice vector
Glide plane	$\det = -1$	fractional

Why T is Abelian Normal?

- $T = \{ \{E | \mathbf{t}\} | \mathbf{t} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3, n_i \in \mathbb{Z} \}$ is a subgroup of the space group.
- $\{E | \mathbf{t}_1\} \{E | \mathbf{t}_2\} = \{E | \mathbf{t}_1 + \mathbf{t}_2\}$
- $\{R | \mathbf{v}\}^{-1} = \{R^{-1} | -R^{-1} \mathbf{v}\}$.
- Compute $\{R | \mathbf{v}\} \{E | \mathbf{t}\} \{R | \mathbf{v}\}^{-1} = \{R | \mathbf{v} + R\mathbf{t}\} \{R^{-1} | -R^{-1} \mathbf{v}\} = \{E | R\mathbf{t}\}$

Two key consequences

- 1 Result is a **pure translation** $\Rightarrow T$ is normal in G ✓
- 2 The translation vector **rotates**: $\mathbf{t} \mapsto R\mathbf{t}$
This requires $R\mathbf{t} \in T$, i.e. R must be a symmetry of the lattice.

Recall: Direct Product vs Semi-direct Product

Direct product $T \times Q$

$$(t_1, R_1)(t_2, R_2) = (t_1 t_2, R_1 R_2)$$

- Both T and Q normal
- Components multiply *independently*
- $tR = Rt$ for all t, R

Semi-direct product $T \rtimes Q$

$$(t_1, R_1)(t_2, R_2) = (t_1 + R_1 t_2, R_1 R_2)$$

- Only T normal
- Q acts on T via $\varphi_R(t) = Rt$
- $tR \neq Rt$ in general

Why not direct?

Translations and rotations *do not commute*: applying R then t gives $R\mathbf{x} + t$, but t then R gives $R(\mathbf{x} + t) = R\mathbf{x} + Rt \neq R\mathbf{x} + t$.

What are Space Groups?

Space groups G , are the **group extensions** of Q (**point groups**) by T (**translational subgroup**). With $T \trianglelefteq G$ (abelian normal subgroup), and $G/T \cong Q$. In 2D the group extension provides 17 groups called the **wallpaper groups**.

$$1 \rightarrow T \rightarrow G \rightarrow Q \rightarrow 1$$

- ① **Symmorphic space groups:** Extension splits, so $G = T \rtimes Q$
- ② **Non-symmorphic space groups:** Extension doesn't split. Replacing some point operations in Q by screw axes and glide planes.

History

First complete list of **230** space groups made by **Fedorov** (Russia, 1891), and independently by **Schoenflies** (Germany, 1891) by purely geometric case analysis. **Zassenhaus** (Germany, 1948) provided the foundational algebraic treatment, with an algorithm to find all space groups.

Symmorphic vs Non-symmorphic

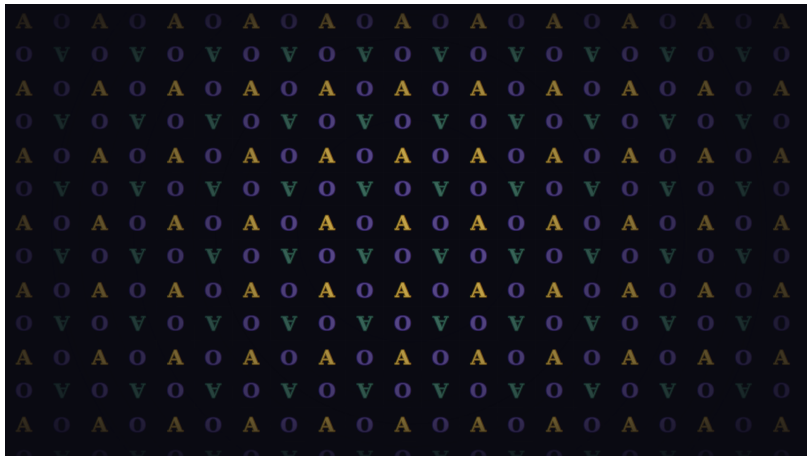
	Symmorphic	Non-symmorphic
Structure	$G = T \rtimes Q$	No clean decomposition
Point ops in G	Pure: $\{R \mathbf{0}\}$ exist	Only $\{R \mathbf{f}\}$, $\mathbf{f}$ fractional
Q subgroup of G	Yes	No
Extra operations	None	Screw axes, glide planes
Count	73	157

Key insight for non-symmorphic

For a screw $\{R | \mathbf{f}\}$ of order n : $\{R | \mathbf{f}\}^n = \{E | \sum_{k=0}^{n-1} R^k \mathbf{f}\} \in T$, so $\mathbf{f}$ is *quantized*: $\mathbf{f} = \frac{p}{n} \mathbf{a}$, $p = 1, \dots, n-1$. Pure fractional translations $\{E | \mathbf{f}\}$ **never** appear in G .

Glide reflection

A reflection σ , followed by a fractional translation $\mathbf{f} = \frac{1}{2}\mathbf{a}$



Screw Axes: Complete Classification

From $n \in \{2, 3, 4, 6\}$ and $\mathbf{f} = \frac{p}{n}\mathbf{a}$:

n	Screws
2	2_1
3	$3_1, 3_2$
4	$4_1, 4_2, 4_3$
6	$6_1, 6_2, 6_3, 6_4, 6_5$
Total	11 screw axes

The allowed values of p are fixed by two constraints: (1) crystallographic restriction on n , and (2) $n\mathbf{f}$ must equal a lattice vector.

Group Cohomology Classification

A space group G is a **group extension** of Q by T :

$$1 \longrightarrow T \longrightarrow G \longrightarrow Q \longrightarrow 1$$

2-cocycles $\varphi : Q \times Q \rightarrow T$ encode the fractional translations; two cocycles give equivalent space groups iff they differ by a **2-coboundary**.

$$H^2(Q, T) = \frac{\{2\text{-cocycles}\}}{\{2\text{-coboundaries}\}}$$

Proof via group cohomology

Space groups with fixed T and Q are classified by

$$H^2(Q, T)$$

Each cohomology class = one inequivalent way to combine T and Q . Summing over all compatible (T, Q) pairs gives exactly **230**.

Representation theory of Space Groups

Groups and their Representations

Definition: Representation

A **representation** of a group G is a homomorphism

$$\rho : G \longrightarrow GL(V)$$

where V is a vector space over $\mathbb{R}$.

- $\rho(e) = E$
- $\rho(g_1 g_2) = \rho(g_1) \rho(g_2)$

Redundancies

- 1 $\rho^{(1)}, \rho^{(2)} = P \rho^{(1)} P^{-1}$, can be removed by choosing unitary matrices.
- 2 $\rho^{(3)} = \begin{bmatrix} \rho^{(1)} & \\ & \rho^{(2)} \end{bmatrix}$, can be removed by focusing on irreducible representations (**irreps**).

Orthogonality and direct product representations

Great Orthogonality Theorem

$$\sum_{g \in G} \left[\rho^{(\mu)}(g) \right]_{ij}^* \left[\rho^{(\nu)}(g) \right]_{kl} = \frac{|G|}{d_\mu} \delta_{\mu\nu} \delta_{ik} \delta_{jl}$$

where d_μ is the dimension of μ -th irrep vector space.

Inner-direct product representation

If $G = H \otimes K$, and $\rho_H^{(i)}$'s, $\rho_K^{(j)}$'s are the representations of subgroups. Then the representation of G , $\rho_G^{(ij)} = \rho_H^{(i)} \otimes \rho_K^{(j)}$, where the last $\otimes$ means the Kronecker matrix product.

Character $\chi^{(\mu)}(g) = \text{Tr } \rho^{(\mu)}(g)$ depends only on conjugacy class, and the number of irreps is equal to the number of conjugacy classes. Key identity: $\sum_{\mu} d_{\mu}^2 = |G|$.

Schur's Lemma and Decomposition into irreps

Schur's Lemma

Let $\rho : G \rightarrow GL(V)$ and $\sigma : G \rightarrow GL(W)$ be two irreps. If $T : V \rightarrow W$ is a linear map satisfying

$$T \rho(g) = \sigma(g) T \quad \forall g \in G,$$

then either $T = 0$, or T is an isomorphism (and $\rho \cong \sigma$).

Every representation ρ decomposes uniquely:

Example:

$$\rho \cong \bigoplus_{\mu} n_{\mu} \rho^{(\mu)}, \quad n_{\mu} = \frac{1}{|G|} \sum_g \chi(g)^* \chi^{(\mu)}(g)$$

$$\rho(g) = \begin{bmatrix} \rho^{(1)}(g) & & \\ & \rho^{(2)}(g) & \\ & & \rho^{(2)}(g) \end{bmatrix}$$

Irreps of the Translation Group

- The translation subgroup T is abelian:

$$T = \{\{E | \mathbf{t}\} | \mathbf{t} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3, N_i > n_i \in \mathbb{Z}^+\}$$

- T can be decomposed as inner-direct product $T = T_1 \otimes T_2 \otimes T_3$ of cyclic groups of order N_1, N_2 , and N_3 .
- Since the cyclic groups are abelian, only one-dimensional representations, and # of irreps $= |T_i| = N_i$.

$$\rho^{(p_i)}(\{E | n_i \mathbf{a}_i\}) = \exp(-2\pi i n_i p_i / N_i)$$

- Calling, $k_i = p_i / N_i$, we can then define

$$\mathbf{k} = k_1 \mathbf{g}_1 + k_2 \mathbf{g}_2 + k_3 \mathbf{g}_3$$

where $\mathbf{a}_i \cdot \mathbf{g}_j = \delta_{ij}$.

Irreps of the Translation Group contd.

- Using the $\mathbf{k}$ label, we can label the irreps of T as

$$\rho^{\mathbf{k}}(\{E \mid n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3\}) = \exp\{-i\mathbf{k} \cdot (n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3)\}$$

- Each irrep is labelled by a wavevector $\mathbf{k}$ in the **first Brillouin zone**

$$\rho^{\mathbf{k}'}(\mathbf{t}) = \rho^{\mathbf{k}}(\mathbf{t}), \quad \text{with } \mathbf{k}' = \mathbf{k} + \mathbf{g}$$

where $\mathbf{g}$ is the reciprocal lattice vector, $\mathbf{g} = m_1\mathbf{g}_1 + m_2\mathbf{g}_2 + m_3\mathbf{g}_3$, $m_i \in \mathbb{Z}$.

Induced and Little Group Representations

Little group (stabilizer) $G_{\mathbf{k}}$

For wavevector $\mathbf{k}$, the **little group** is

$$G_{\mathbf{k}} = \{g \in G \mid g\mathbf{k} = \mathbf{k} + \mathbf{g}, \mathbf{g} \text{ reciprocal lattice vector}\}$$

Its irreps label the energy bands at $\mathbf{k}$.

Induced representation

An irrep $\rho^{(\mu)}$ of $G_{\mathbf{k}}$ **induces** a representation of the full group G :

$$\text{Ind}_{G_{\mathbf{k}}}^G \rho^{(\mu)}$$

This describes how bands transform as $\mathbf{k}$ moves through the Brillouin zone.

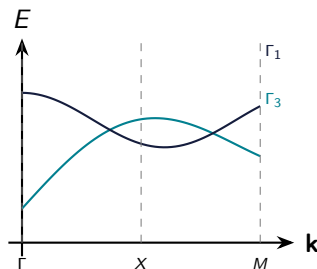
Bandstructure and consequences of symmetries

Band Structure as Irrep Decomposition

At each $\mathbf{k}$, the Hilbert space decomposes into irreps of $G_{\mathbf{k}}$:

$$\mathcal{H}_{\mathbf{k}} = \bigoplus_{\mu} \bigoplus_{n: \text{band}} V^{(\mu)}$$

- Degeneracy at $\mathbf{k} = d_{\mu}$ (dimension of irrep)
- Band sticking along a line: irreps of the group
- Compatibility relations: how irreps at high-symmetry points connect along paths



Symmetry of the Hamiltonian (G –invariant Hamiltonians)

Definition. G is a symmetry group of Hamiltonian H if every $g \in G$ acts on $\mathcal{H}$ by a unitary operator $U(g)$ satisfying

$$[H, U(g)] = 0 \quad \forall g \in G.$$

Step 1: Symmetries map eigenspaces to eigenspaces

Let $H|\psi\rangle = E|\psi\rangle$. Apply $U(g)$:

$$H(U(g)|\psi\rangle) = U(g)H|\psi\rangle = E(U(g)|\psi\rangle).$$

$U(g)|\psi\rangle$ is now an eigenstate with *the same energy* E . The eigenspace $V_E = \{|\psi\rangle \in \mathcal{H} : H|\psi\rangle = E|\psi\rangle\}$ is a subspace of Hilbert space which is **G -invariant**, i.e., **action of G stays within V_E** .

Consequence

Restricting $U(g)$ to V_E gives a unitary representation $\rho_E : G \rightarrow GL(V_E)$.

Wigner's theorem

Wigner's theorem

The labels for irreps of the symmetry group are "good quantum numbers" and can be used to label quantum states. From the irrep decomposition of the group, the degeneracy could be determined, and that irrep we use as the labels. Let H be a Hamiltonian with symmetry group G . Each energy eigenspace V_E decomposes as a direct sum of irreps of G . For a *generic* Hamiltonian with symmetry exactly G (no accidental degeneracy), V_E is itself a **single irrep** V^μ , and

$$\text{degeneracy of } E = \dim V^\mu = d_\mu.$$

Degeneracies

The degeneracies fall under two categories: **symmetry-protected degeneracies** (these degeneracies cannot be removed by putting symmetry-protected perturbations) and **accidental degeneracies** (in most cases, they could be removed by small perturbations to the Hamiltonian)

Summary

Summary

- 1 **14 Bravais lattices** classify translation subgroups T .
- 2 **32 point groups** are forced by the crystallographic restriction $2 \cos \theta \in \mathbb{Z}$.
- 3 For **symmorphic** space groups: $G = T \rtimes Q$ (semi-direct, not direct, because $\{R | \mathbf{0}\}\{E | \mathbf{t}\} \neq \{E | \mathbf{t}\}\{R | \mathbf{0}\}$).
- 4 For **non-symmorphic** space groups: $G \neq T \rtimes Q$; point operations carry fractional translations (screws, glides).
- 5 All 230 space groups are classified by $H^2(Q, T)$ over all compatible (T, Q) pairs.
- 6 $[H, U(g)] = 0 \Rightarrow V_E$ is G -invariant, and irreps of G could be used as state labels.
- 7 These state labels sometimes provide insights into the state transitions (**selection rules**).
- 8 Incorporating **topological invariants** of bands with space groups to identify and classify new classes of materials. Bradlyn, Jennifer Cano, et al. **Topological quantum chemistry**, Nature volume 547, pages298–305 (2017).

Thank you!

Bloch's Theorem:



Symmorphic Groups example: $P6/mmm$



Example: Point Groups of Monoclinic crystals

